

Spring Data JPA - Quick Example and Difference between JPA, Hibernate and Spring Data JPA

Spring Data JPA – Quick Example

Spring Data JPA simplifies database operations in Spring Boot by providing ready-made repository methods. A typical flow is:

MySQL Database → Country Table → Entity (Country.java) → Repository (JpaRepository) → Service Layer → Spring Boot Application → Output.

Example:

1. Create a Country entity using @Entity.
2. Create CountryRepository extending JpaRepository.
3. Create CountryService and call repository.findAll().
4. Invoke the service from the Spring Boot application.

Expected Output:

IN India

US United States

Difference between JPA, Hibernate and Spring Data JPA

Feature	JPA	Hibernate	Spring Data JPA
What is it?	Specification	ORM Framework	Framework built on JPA
Implementation	No	Yes	No
Purpose	Defines persistence rules	Implements JPA	Reduces boilerplate code
CRUD	Manual	Manual	Automatic using JpaRepository
Transactions	Specification	Manual	Easy with @Transactional
Best Used	Standard API	ORM implementation	Spring Boot applications

Relationship

Application → Service → Repository (Spring Data JPA) → Hibernate → Database

Key Points

- JPA is a specification and does not contain an implementation.
- Hibernate is an ORM framework that implements JPA.
- Spring Data JPA works on top of Hibernate (or another JPA provider).
- JpaRepository provides built-in CRUD methods such as save(), findAll(), findById() and deleteById().
- Spring Data JPA reduces boilerplate code and speeds up development.